Fundamentals Chapter 13

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1. Let $q \in \mathbb{Q}$ be any upper bound of $S = \{1, 1.4, 1.41, \ldots\}$. We wish to find $q' \in \mathbb{Q}$ such that $s \leq q' < q$ for all $s \in S$.

Since $\sqrt{2}$ is irrational, $q \neq \sqrt{2}$ and $q > s$ for all $s \in S$. From problem 4a, we know that there exists an integer $m > 0$ such that $\frac{1}{m} < q - \sqrt{2}$. Let $q' = \sqrt{2} + \frac{1}{m}$. Since $0 < \frac{1}{m} < q - \sqrt{2}$, we know that

$$\sqrt{2} < \sqrt{2} + \frac{1}{m} = q' < \sqrt{2} + (q - \sqrt{2}) = q,$$

or $\sqrt{2} < q' < q$.

2. $\mathbb{N}$ and $\mathbb{Z}$ are not fields and $\mathbb{C}$ is not a totally ordered set.

3a. We must show the existence of an identity, the existence of an inverse, and associativity. The identity would be the sequence (e_n) where $e_n = 0$ for all n, since $(x_n) + (e_n) = (x_n + e_n) = (0)$. The inverse of (x_n) would be $y_n = -x_n$, since $(x_n) + (y_n) = (x_n - x_n) = (0)$.

We have that for all $k \in \mathbb{N}$, $(a_k + b_k) + c_k = a_k + (b_k + c_k)$, so $((a_n) + (b_n)) + (c_n) = (a_n) + ((b_n) + (c_n))$, so associativity holds.

3b. We need to show that $C/H = C/\sim$. We have that

$$C/H = \{(a_n) + H : (a_n) \in C\} = \{\{(a_n) + (b_n) : (b_n) \to 0\} : (a_n) \in C\},$$

$$C/\sim = \{\{(b_n) : (a_n) \sim (b_n)\} : (a_n) \in C\} = \{\{(b_n) : (a_n) - (b_n) \to 0\} : (a_n) \in C\}.$$

So to show that $C/H = C/\sim$, we need to show that

$$\{(a_n) + (b_n) : (b_n) \to 0\} = \{(b_n) : (a_n) - (b_n) \to 0\}$$

given any $(a_n) \in C$. We have that $(a_n) - (b_n) \to 0$ iff $(b_n) - (a_n) \to 0$, so we can show that

$$\{(a_n) + (b_n) : (b_n) \to 0\} = \{(b_n) : (b_n) - (a_n) \to 0\}.$$

We can define $(c_n) = (b_n) - (a_n)$, so the RHS becomes

$$\{(a_n) + (c_n) : (c_n) \to 0\},$$

which is equal to the LHS.

4a. We have that $x = [(x_n)]$ for some Cauchy sequence (x_n). Since $(x_n) \not\to 0$, there exists N such that if $n > N$ then $x_n \neq 0$. Define (y_n) so that $y_n = \frac{1}{x_{n+N}}$. Since (x_{n+N}) is Cauchy and never touches 0, y_n is bounded. So choose an integer $m > \sup y_n$. Then $m > \frac{1}{x_{n+N}}$ for all $n \in \mathbb{N}$, so $x_{n+N} > \frac{1}{m}$, and therefore $x > \frac{1}{m}$.

4b. Choose m such that $\frac{1}{m} < \frac{y-x}{2}$. Then

$$x + \frac{1}{m} < x + \frac{y-x}{2} = \frac{x+y}{2},$$

$$y - \frac{1}{m} > y - \frac{y-x}{2} = y + \frac{x-y}{2} = \frac{x+y}{2}.$$

So $x + \frac{1}{m} < \frac{x+y}{2} < y - \frac{1}{m}$.

4c. Assume $x = 0$. Then clearly $|x| < \varepsilon$ for all rational $\varepsilon > 0$.

Now assume $x < \varepsilon$ for all rational $\varepsilon > 0$, and that $x \neq 0$. Choose $\varepsilon = \frac{1}{m} < |x|$ for some positive integer m. Then $|x| > \varepsilon$, which is a contradiction, so $x = 0$.

5a. The greatest lower bound of a set S is the real number L such that $L' \leq L \leq s$ for all $s \in S$ and all lower bounds L' of S.

5b. Since b is the least upper bound of S, we have that $L \geq b \geq s$ for all $s \in S$ and upper bounds L of S. This means $-L \leq -b \leq -s$ as well. We know that $-s \in -S$, so $-L$ is always a lower bound of $-S$. By the definition of greatest lower bound, $-b$ is the greatest lower bound of $-S$.

5c. Take any subset S of $C/\sim$. We know that $-S$ has a least upper bound, so S has a greatest lower bound.

6. We know that a_1 is not an upper bound for S. Also, since a_{n+1} is defined as a_n if c_1 is an upper bound and c_1 otherwise, if a_n is not an upper bound for S then a_{n+1} is not. By induction, none of the terms of (a_n) are upper bounds of S. So if v is an upper bound then $a_n < v$ for all n, so $[(a_n)] \leq v$.

7a. Take any m, n such that $m > n$. We have that $S_m \subseteq S_n$ so any element of S_m is in S_n. This means the smallest element of S_m is also in S_n, so $\inf S_m \geq \inf S_n$, so $(\inf S_m)$ is monotone increasing.

Similarly, take m, n such that $m > n$. We have that $S_m \subseteq S_n$, so the largest element of S_m is also in S_n, meaning $\sup S_m \leq \sup S_n$. Therefore, $(\sup S_m)$ is

monotone decreasing.

7b. We say that $x_n \to \infty$ if for all integers M there exists N such that $x_n > M$ whenever $n > N$ and $x_n \to -\infty$ if for all integers M there exists N such that $x_n < M$ whenever $n > N$.

7c. If $\liminf x_n$ or $\limsup x_n$ is bounded, then since they are monotone, they converge to a real number. If they are not bounded, then they diverge to ∞ or $-\infty$.

7d. Assume $\lim x_n$ exists and call it L. Given $\varepsilon > 0$, there is N such that for all $n > N$, $|x_n - L| < \varepsilon$. So for all $m \geq N$, every x_n with $n \geq m$ is within ε of L. So

$$\inf\{x_n : n \geq m\} > L - \varepsilon \quad \text{and} \quad \sup\{x_n : n \geq m\} < L + \varepsilon.$$

Taking $m \to \infty$, we see that

$$\lim_{m\to\infty} \inf S_m \geq L - \varepsilon, \quad \lim_{m\to\infty} \sup S_m \leq L + \varepsilon.$$

Therefore, both limits are exactly L.

Now, assume $\liminf x_n = \limsup x_n = L$. We know that for all $\varepsilon > 0$ there is N_1 such that for all $m > N_1$, $\inf S_m > L - \varepsilon$ and there is N_2 such that for all $m > N$, $\sup S_m < L + \varepsilon$. So, if $n > N$ where $N = \max(N_1, N_2)$, we have that

$$L - \varepsilon < \inf S_m \leq x_n \leq \sup S_m < L + \varepsilon,$$

so $|x_n - L| < \varepsilon$. Therefore, $x_n \to L$.

8. $x_n = n(-1)^n$.

9. Since $\inf S_m = -1$ and $\sup S_m = 1$ regardless of how large m is, $\liminf x_n = -1$ and $\liminf x_n = 1$, so $x_n = (1, -1, 1, ...)$ doesn't converge.

10. Define x^y as $[(c_n)]$ where $c_n = a_n^{b_n}$ for all n.